Algebra 1 — Comprehensive Review Worksheet

Factoring · Completing the Square · Graphing Quadratics · Sequences · Exponential Transformations

Name: _____ **Date:** _____ **Period:** _____ **Score:** _____ / 50

Section 1: Solving Equations by Factoring

Solve: $x^2 + 5x + 6 = 0$

Step 1. Find two numbers that multiply to 6 and add to 5 $\longrightarrow$ 2 and 3
Step 2. Factor: $(x+2)(x+3) = 0$
Step 3. Zero Product Property: $x + 2 = 0 \Rightarrow \boldsymbol{x = -2}$ $x + 3 = 0 \Rightarrow \boldsymbol{x = -3}$

A. Multiple Choice

(Circle the best answer.)

1. Solve: $x^2 + 7x + 12 = 0$
a) $x = 3,\ 4$ b) $x = -3,\ -4$ c) $x = -3,\ 4$ d) $x = 3,\ -4$

2. Solve: $x^2 - 5x + 6 = 0$
a) $x = -2,\ -3$ b) $x = 2,\ 3$ c) $x = -2,\ 3$ d) $x = 2,\ -3$

3. Solve: $x^2 - 9 = 0$
a) $x = 9$ b) $x = 3$ c) $x = \pm 3$ d) $x = \pm 9$

4. Solve: $2x^2 + 5x + 2 = 0$
a) $x = -\frac{1}{2},\ -2$ b) $x = \frac{1}{2},\ 2$ c) $x = -\frac{1}{2},\ 2$ d) $x = \frac{1}{2},\ -2$

5. Solve: $x^2 - x - 12 = 0$
a) $x = -4,\ 3$ b) $x = 4,\ -3$ c) $x = 4,\ 3$ d) $x = -4,\ -3$

B. Practice Problems

(Show all work.)

1. Solve: $x^2 + 8x + 15 = 0$

Work space

2. Solve: $x^2 - 6x + 8 = 0$

Work space

3. Solve: $x^2 - 25 = 0$

Work space

4. Solve: $3x^2 + 7x + 2 = 0$

Work space

5. Solve: $x^2 + 2x - 15 = 0$

Work space

Section 2: Solving Equations by Completing the Square

Solve: $x^2 + 6x + 5 = 0$

Step 1. Move constant: $x^2 + 6x = -5$

Step 2. Add $\left(\frac{6}{2}\right)^2 = 9$ to both sides: $x^2 + 6x + 9 = 4$

Step 3. Perfect square: $(x + 3)^2 = 4$

Step 4. Square root: $x + 3 = \pm 2 \Rightarrow \boldsymbol{x = -1}$ or $\boldsymbol{x = -5}$

A. Multiple Choice

(Circle the best answer.)

1. Solve by completing the square: $x^2 + 4x - 5 = 0$

a) $x = 1,\ -5$ b) $x = -1,\ 5$ c) $x = 1,\ 5$ d) $x = -1,\ -5$

2. Solve: $x^2 - 6x + 5 = 0$

a) $x = -1,\ -5$ b) $x = 1,\ 5$ c) $x = 2,\ 3$ d) $x = -2,\ -3$

3. What number completes the square for $x^2 + 10x + _____$?

a) 5 b) 10 c) 25 d) 100

4. Solve: $x^2 - 8x + 7 = 0$

a) $x = 1,\ 7$ b) $x = -1,\ -7$ c) $x = 2,\ 4$ d) $x = -2,\ -4$

5. Solve: $x^2 + 2x - 3 = 0$

a) $x = -3,\ 1$ b) $x = 3,\ -1$ c) $x = 3,\ 1$ d) $x = -3,\ -1$

B. Practice Problems

(Show all work.)

1. Solve by completing the square: $x^2 + 6x + 8 = 0$

Work space

2. Solve by completing the square: $x^2 - 4x - 12 = 0$

Work space

3. Solve by completing the square: $x^2 + 8x + 12 = 0$

Work space

4. Solve by completing the square: $x^2 - 10x + 16 = 0$

Work space

5. Solve by completing the square: $x^2 + 12x + 20 = 0$

Work space

Section 3: Graphing Quadratic Equations

Graph $y = x^2 - 4x + 3$ and identify all key features.

Axis of Symmetry: $x = -\dfrac{b}{2a} = -\dfrac{-4}{2(1)} = 2$

Vertex: $y = (2)^2 - 4(2) + 3 = -1 \Rightarrow$ **Vertex** $= (2, -1)$
y-intercept: $(0, 3)$ **x-intercepts:** $(x-1)(x-3) = 0 \Rightarrow x = 1,\ x = 3$
Domain: $-\infty < x < \infty$ **Range:** $y \geq -1$

A. Multiple Choice

(Circle the best answer.)

1. For $y = x^2 - 6x + 5$, what is the vertex?
a) $(3, -4)$ b) $(-3, 4)$ c) $(3, 4)$ d) $(6, 5)$

2. What is the axis of symmetry for $y = 2x^2 + 8x - 1$?
a) $x = 4$ b) $x = -4$ c) $x = -2$ d) $x = 2$

3. What is the range of $y = x^2 + 2$?
a) $y \geq 0$ b) $y \geq 2$ c) all reals d) $y \leq 2$

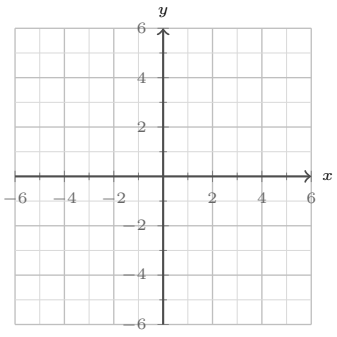
4. Which parabola opens downward?
a) $y = 3x^2$ b) $y = x^2 - 5$ c) $y = -2x^2 + 1$ d) $y = (x-1)^2$

5. The domain of ANY quadratic $y = ax^2 + bx + c$ is:
a) $x \geq 0$ b) $x \leq 0$ c) all reals d) depends on a

B. Practice Problems

(Graph each parabola and identify all key features.)

1. $y = x^2 - 2x - 3$



Vertex: _____

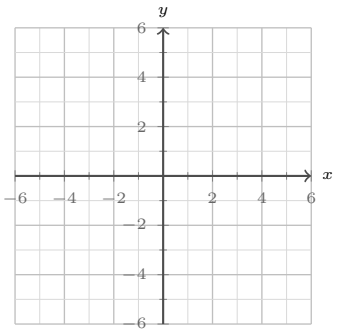
Axis of Symmetry: _____

Domain: _____

Range: _____

Opens: Up Down

2. $y = -x^2 + 4x$



Vertex: _____

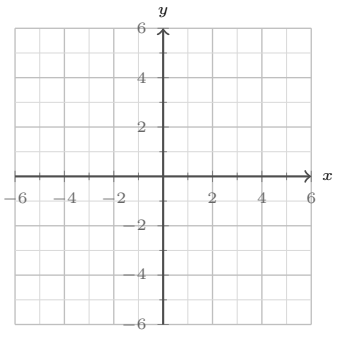
Axis of Symmetry: _____

Domain: _____

Range: _____

Opens: Up Down

3. $y = x^2 + 4x + 4$



Vertex: _____

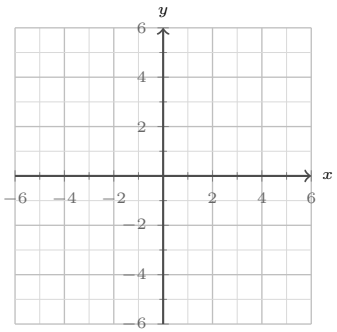
Axis of Symmetry: _____

Domain: _____

Range: _____

Opens: Up Down

4. $y = 2x^2 - 8x + 6$



Vertex: _____

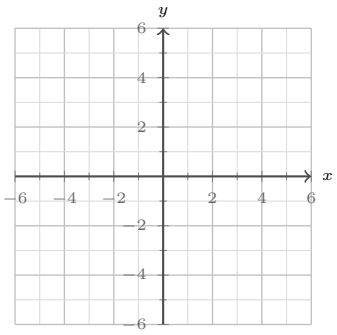
Axis of Symmetry: _____

Domain: _____

Range: _____

Opens: Up Down

5. $y = -x^2 + 6x - 5$



Vertex: _____

Axis of Symmetry: _____

Domain: _____

Range: _____

Opens: Up Down

Section 4: Arithmetic & Geometric Sequences — The nth Term

Arithmetic (add constant difference d): $a_n = a_1 + (n-1)d$
Sequence: 3, 7, 11, 15, ... $a_1 = 3,\ d = 4 \quad \Rightarrow \quad a_n = 3 + (n-1)\cdot 4 = \mathbf{4n - 1}$

Geometric (multiply by constant ratio r): $a_n = a_1 \cdot r^{n-1}$
Sequence: 2, 6, 18, 54, ... $a_1 = 2,\ r = 3 \quad \Rightarrow \quad a_n = \mathbf{2 \cdot 3^{n-1}}$

A. Multiple Choice

(Circle the best answer.)

1. Find the 10th term of the arithmetic sequence: 5, 9, 13, 17, ...
a) 40 b) 41 c) 45 d) 36

2. Find the nth term formula for: 2, 5, 8, 11, ...
a) $a_n = 3n - 1$ b) $a_n = 3n + 2$ c) $a_n = 2n + 1$ d) $a_n = n + 3$

3. Find the 5th term of the geometric sequence: 3, 6, 12, 24, ...
a) 48 b) 96 c) 36 d) 192

4. Which sequence is geometric?
a) 2, 4, 6, 8 b) 1, 3, 6, 10 c) 4, 12, 36, 108 d) 5, 10, 15, 20

5. The formula $a_n = 2 \cdot 4^{n-1}$ describes which type of sequence?
a) Arithmetic, $d = 4$ b) Geometric, $r = 4$ c) Geometric, $r = 2$ d) Arithmetic, $d = 2$

B. Practice Problems

(Show all work.)

1. Write the nth term formula and find a_{12} for: 7, 11, 15, 19, ...

Work space

2. Write the nth term formula and find a_6 for the geometric sequence: 5, 15, 45, 135, ...

Work space

3. Find the 20th term: -3, 1, 5, 9, ...

Work space

4. A geometric sequence has $a_1 = 100$ and $r = \frac{1}{2}$. Write the nth term formula and find a_5.

Work space

5. Is the sequence 64, 32, 16, 8, ... arithmetic or geometric? Write the nth term formula.

Work space

Section 5: Transformations of Exponential Functions

Parent function: $f(x) = 2^x$ **General form:** $g(x) = a \cdot b^{x-h} + k$

$\lvert a\rvert > 1$	vertical stretch
$0 < \lvert a\rvert < 1$	vertical compression
$a < 0$	reflection over the x-axis
replace x with $-x$	reflection over the y-axis
$h > 0$	shift right h units
$h < 0$	shift left $\lvert h\rvert$ units
$k > 0$	shift up k units
$k < 0$	shift down $\lvert k\rvert$ units

Example: $g(x) = -3 \cdot 2^{x+2} - 1 \Rightarrow$ reflect over x-axis, vertical stretch $\times 3$, left 2, down 1. Asymptote: $y = -1$

A. Multiple Choice

(Circle the best answer.)

1. How does $g(x) = 2^x + 5$ compare to $f(x) = 2^x$?

a) Shift left 5 b) Shift right 5 c) Shift up 5 d) Shift down 5

2. Describe the transformation: $g(x) = 2^{x-3}$

a) Shift up 3 b) Shift down 3 c) Shift left 3 d) Shift right 3

3. Which equation represents a vertical stretch by a factor of 4 of $f(x) = 3^x$?

a) $f(x) = 3^{4x}$ b) $f(x) = 4 \cdot 3^x$ c) $f(x) = 3^x + 4$ d) $f(x) = \frac{1}{4} \cdot 3^x$

4. What is the horizontal asymptote of $g(x) = 2^x - 7$?

a) $y = 2$ b) $y = 7$ c) $y = -7$ d) $y = 0$

5. $g(x) = -2^x$ is a reflection of $f(x) = 2^x$ over which axis?

a) y-axis b) x-axis c) the origin d) the line $y = x$

B. Practice Problems

(Describe all transformations fully.)

1. Describe all transformations: $g(x) = 3 \cdot 2^{x-1} + 4$

Work space

2. Write the equation of $f(x) = 4^x$ shifted **left 2**, reflected over the x-axis, and shifted **down 3**.

Work space

3. State the domain, range, and asymptote of $g(x) = -2 \cdot 5^x + 1$

Work space

4. Describe all transformations: $g(x) = \frac{1}{2} \cdot 3^{-x} - 2$

Work space

5. The graph of $f(x) = 2^x$ is transformed to pass through $(0, 6)$ with asymptote $y = -1$. Write a possible equation in the form $g(x) = a \cdot 2^x + k$.

Work space

Answer Key

Total: 50 points | Multiple Choice: 25 pts (1 pt each) | Practice: 25 pts (1 pt each)

Section 1 — Solving by Factoring

Multiple Choice: 1. B 2. B 3. C 4. A 5. B

Practice:

1. $x^2 + 8x + 15 = 0 \Rightarrow (x+3)(x+5) = 0 \Rightarrow \boldsymbol{x = -3,\ -5}$
2. $x^2 - 6x + 8 = 0 \Rightarrow (x-2)(x-4) = 0 \Rightarrow \boldsymbol{x = 2,\ 4}$
3. $x^2 - 25 = 0 \Rightarrow (x-5)(x+5) = 0 \Rightarrow \boldsymbol{x = \pm 5}$
4. $3x^2 + 7x + 2 = 0 \Rightarrow (3x+1)(x+2) = 0 \Rightarrow \boldsymbol{x = -\frac{1}{3},\ -2}$
5. $x^2 + 2x - 15 = 0 \Rightarrow (x+5)(x-3) = 0 \Rightarrow \boldsymbol{x = -5,\ 3}$

Section 2 — Completing the Square

Multiple Choice: 1. A 2. B 3. C 4. A 5. A

Practice:

1. $x^2 + 6x + 8 = 0 \Rightarrow (x+3)^2 = 1 \Rightarrow \boldsymbol{x = -2,\ -4}$
2. $x^2 - 4x - 12 = 0 \Rightarrow (x-2)^2 = 16 \Rightarrow \boldsymbol{x = 6,\ -2}$
3. $x^2 + 8x + 12 = 0 \Rightarrow (x+4)^2 = 4 \Rightarrow \boldsymbol{x = -2,\ -6}$
4. $x^2 - 10x + 16 = 0 \Rightarrow (x-5)^2 = 9 \Rightarrow \boldsymbol{x = 8,\ 2}$
5. $x^2 + 12x + 20 = 0 \Rightarrow (x+6)^2 = 16 \Rightarrow \boldsymbol{x = -2,\ -10}$

Section 3 — Graphing Quadratics

Multiple Choice: 1. A 2. C 3. B 4. C 5. C

Practice:

1. $y = x^2 - 2x - 3$: Vertex $\boldsymbol{(1, -4)}$, AOS $x = 1$, Domain: all reals, Range: $y \geq -4$, Opens up
2. $y = -x^2 + 4x$: Vertex $\boldsymbol{(2, 4)}$, AOS $x = 2$, Domain: all reals, Range: $y \leq 4$, Opens down
3. $y = x^2 + 4x + 4$: Vertex $\boldsymbol{(-2, 0)}$, AOS $x = -2$, Domain: all reals, Range: $y \geq 0$, Opens up
4. $y = 2x^2 - 8x + 6$: Vertex $\boldsymbol{(2, -2)}$, AOS $x = 2$, Domain: all reals, Range: $y \geq -2$, Opens up
5. $y = -x^2 + 6x - 5$: Vertex $\boldsymbol{(3, 4)}$, AOS $x = 3$, Domain: all reals, Range: $y \leq 4$, Opens down

Section 4 — Sequences

Multiple Choice: 1. B 2. A 3. A 4. C 5. B

Practice:

1. $d = 4$; $a_n = 4n + 3$; $\boldsymbol{a_{12} = 51}$

2. $r = 3$; $a_n = 5 \cdot 3^{n-1}$; $\boldsymbol{a_6 = 1215}$
3. $d = 4$; $a_n = 4n - 7$; $\boldsymbol{a_{20} = 73}$
4. $a_n = 100 \cdot \left(\frac{1}{2}\right)^{n-1}$; $\boldsymbol{a_5 = 6.25}$
5. **Geometric**, $r = \frac{1}{2}$; $a_n = 64 \cdot \left(\frac{1}{2}\right)^{n-1}$

Section 5 — Exponential Transformations

Multiple Choice: 1. C 2. D 3. B 4. C 5. B

Practice:

1. Vertical stretch $\times 3$, shift right 1, shift up 4
2. $\boldsymbol{g(x) = -4^{x+2} - 3}$
3. Domain: all reals; Range: $y < 1$; Asymptote: $\boldsymbol{y = 1}$
4. Vertical compression $\times \frac{1}{2}$, reflection over the y-axis, shift down 2
5. $g(0) = a + k = 6$ and $k = -1 \Rightarrow a = 7$; $\boldsymbol{g(x) = 7 \cdot 2^x - 1}$

— End of Worksheet —